Part 3

3.1. Top 5 results

Query 1: illinois

Rank	ID	Title
1	11192	Illinois
2	219587	United States
3	10193	Chicago
4	144	City
5	13404	Democratic Party (United States)

Query 2: Computer Science

Rank	ID	Title
1	700	Science
2	112	Computer
3	110	Computer Science
4	108	Chemistry
5	17651	Science fiction

Query 3: olympic games

Rank	ID	Title
1	5290	Olympic Games
2	291	France
3	219828	Germany
4	403	Japan
5	27	Australia

Query 4: river france

Rank	ID	Title
1	291	France
2	28272	Departments of France
3	219587	United States
4	939	Water
5	219837	World War II

Query 5: music

Rank	ID	Title
1	472	Music
2	219587	United States
3	2041	Pop music
4	939	Water
5	5022	Classical music

```
(base) B99@Bs-MacBook cs511p2 % make query Q="computer science"
 1  700        0.487374     Science
 2  112        0.179018     Computer
 3  110        0.0697992    Computer science
 4  108        0.0411792    Chemistry
 5  17651      0.0380343    Science fiction
 6  502310     0.025062     Human
 7  418926     0.0162321    Natural science
 8  591        0.0129351    Physics
 9  467        0.0128052    Mathematics
10  2957       0.0122785    Computer program
(base) B99@Bs-MacBook cs511p2 % make query Q="illinois"

 1  11192      0.0606128    Illinois
 2  219587     0.0279108    United States
 3  10193      0.0162091    Chicago
 4  144        0.0160571    City
 5  13404      0.00669493   Democratic Party (United States)
 6  368998     0.0037832    2020 United States Census
 7  5414       0.00371866   Abraham Lincoln
 8  4546       0.00340624   Ronald Reagan
 9  223430     0.00311566   Barack Obama
10  884        0.00291881   University
(base) B99@Bs-MacBook cs511p2 % make query Q="olympic games"

 1  5290       0.240989     Olympic Games
 2  291        0.0725688    France
 3  219828     0.0342146    Germany
 4  403        0.0341017    Japan
 5  27         0.0319319    Australia
 6  2964       0.0267132    Russia
 7  2062       0.0246278    Association football
 8  460        0.0227952    London
 9  26435      0.0211033    Summer Olympic Games
10  168777     0.0196931    List of IOC country codes
(base) B99@Bs-MacBook cs511p2 % make query Q="river france"

 1  291        0.974731     France
 2  28272      0.374008     Departments of France
 3  219587     0.188695     United States
 4  939        0.173534     Water
 5  219837     0.0652572    World War II
 6  363        0.0423341    Italy
 7  120        0.0371971    China
 8  856        0.0333386    United Kingdom
 9  219828     0.0323521    Germany
10  144        0.031586     City
(base) B99@Bs-MacBook cs511p2 % make query Q="music"
 1  472        0.225583     Music
 2  219587     0.0589653    United States
 3  2041       0.0433486    Pop music
 4  939        0.0315548    Water
 5  5022       0.0294271    Classical music
 6  12945      0.0247047    Rock and roll
 7  502310     0.0235569    Human
 8  24302      0.0164039    Folk music
 9  566        0.0128533    New York City
10  112        0.0113459    Computer
(base) B99@Bs-MacBook cs511p2 %
```

3.2

Query: university

Rank	ID	Title
1	884	University
2	856	United Kingdom
3	566	New York City
4	3047	England
5	8740	Evolution
6	121093	University of Cambridge
7	219587	United States
8	17333	Philadelphia
9	502310	Human
10	426	Law

The results partially match my expectations. “University” ranks first, and “University of Cambridge” is also relevant. However, other articles like “United Kingdom,” and “New York City” appear above some more directly related university articles

To test PageRank’s impact, I compared the final ranking with a TF-IDF only ranking. “United Kingdom” rises from 311th to 2nd, “New York City” from 195th to 3rd, and “England” from 784th to 4th. As we can see, PageRank significantly helps well-connected articles.

On the other hand, “University of Cambridge” improves from 8th to 6th in the comparison. But its gain is much smaller. As a result, it is overtaken by several general articles.

In conclusion, PageRank helps prioritize globally important pages, but it can sometimes favor general articles over more directly related articles.

```
(base) B99@Bs-MacBook cs511p2 % make query Q="university"
  1  884        0.30052      University
  2  856        0.0573506    United Kingdom
  3  566        0.0365718    New York City
  4  3047       0.0254711    England
  5  8740       0.0247306    Evolution
  6  121093     0.0238142    University of Cambridge
  7  219587     0.0172077    United States
  8  17333      0.0171437    Philadelphia
  9  502310     0.0147312    Human
 10  426        0.00930285   Law
```

Rank	TF-IDF Rank	PageRank	Title
1	4	0.0006746112	University
2	311	0.00214998	United Kingdom
3	195	0.001054628	New York City
4	784	0.001591448	England
5	41	0.0002990677	Evolution
6	8	0.0001159421	University of Cambridge
7	10340	0.006450901	United States
8	42	0.0002073195	Philadelphia
9	1349	0.001104499	Human
10	2389	0.001162495	Law